MIL-T-37497
1 December 1976

MILITARY SPECIFICATION

TRIAMCINOLONE ACETONIDE AEROSOL, NF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Triamcinolone Acetonide Aerosol, NF (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (See 6.3)

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Triamcinolone Acetonide Aerosol in accordance with the tests, standards, and requirements of the NF, including any supplements or revisions thereto. Each gram of formulation shall contain 0.066 mg of Triamcinolone Acetonide USP, 33.6 mg of Isopropyl Palmitate, and 46.3 mg of either Specially Denatured Alcohol 40 or Dehydrated Alcohol, each within the designated assay limits for the preparation. Shall also contain Trichloromonofluoromethane, NF, and Dichlorodifluoromethane NF, as propellants.

3.2 Spray delivery. A three-second continuous spray shall deliver not more than 0.2 mg of Triamcinolone Acetonide when determined by the loss of weight of the immediate container and calculating the amount of Triamcinolone Acetonide based on the assay of the solution.

3.3 Assay.

3.3.1 Triamcinolone Acetonide. The finished solution shall comply with the applicable assay limits for Triamcinolone Acetonide when tested as specified in the NF.

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3.3.2 Isopropyl Palmitate. The finished solution shall assay to contain not less than 3.06 percent and not more than 3.74 percent of Isopropyl Palmitate, when determined as specified in 4.7.2.

3.3.3 Alcohol. The finished aerosol shall assay to contain not less than 2.40 percent (W/W) and not more than 5.20 percent (W/W) of Alcohol (denatured or dehydrated), when determined by bubbling the aerosol sample through distilled water and assaying the solution by the USP method for Alcohol Determination.

3.4 Ingredients.

3.4.1 Triamcinolone Acetonide. The Triamcinolone Acetonide used in the manufacture of the aerosol shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.4.2 Isopropyl Palmitate. The Isopropyl Palmitate used in the preparation of the aerosol shall comply with the following:

3.4.2.1 Ingredient value. Shall have a saponification value between 177 and 198, an iodine value of not more than 4.0, and an acid value of not more than 2.0, when determined according to the USP procedures.

3.4.3 Alcohol. The Alcohol used in the preparation of the aerosol shall conform to the tests, standards, and requirements of the N.F. X monograph for 'Dehydrated Alcohol', which was official from December 15, 1955, to September 30, 1960. In addition, when Specially Denatured Alcohol 40 is used in lieu of Dehydrated Alcohol, it shall be in accordance with Federal regulations.

3.4.4 Other ingredients. All other ingredients entering into the manufacture of the aerosol shall be of USP or NF quality or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.5 Expiration dating period. The expiration dating period shall be 24 months.

3.5.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.6 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of aerosol produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 For raw material and end item testing. Noncompliance with any of the following is cause for rejection.

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TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Spray delivery	3.2	S-2	5
Assay:			
Triamcinolone Acetonide	3.3.1	USP requirements	--
Isopropyl Palmitate	3.3.2	S-2	--
Alcohol	3.3.3	S-2	5
Ingredients:			
Isopropyl Palmitate 2/	3.4.2	--	--

1/ Where possible, the sample shall be used for two or more tests.

2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The finished aerosol shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

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TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Valve does not function properly (continuous or no spray).
102	Valve stem is loose.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.
203	Valve stem not straight.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Tests. Tests shall be performed to determine compliance with all requirements of this document and all referenced documents.

4.7.2 Isopropyl Palmitate (residue). Attach a delivery tube to the actuator button and accurately weigh and record the weight of the can. Invert the can and discharge about 20 to 25 grams into a clean, tared, 100-ml, round bottom flask. Reweigh the can and calculate the weight of sample taken. Add 10 ml of diethyl ether to the flask, mix well, and evaporate on a rotary evaporator, using a water aspirator to create maximum vacuum. Allow the flask to rotate until no more water condenses, add 10 ml of ether and dry on the rotary evaporator for an additional two hours. Continue drying for additional one-hour period until the loss in weight is less than 0.5 percent in an hour. Calculate as follows:

$$\frac{\text{Gram of dry residue} \times 100}{\text{Gram of sample}} = \text{Percent residue (isopropyl Palmitate).}$$

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (see 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One can containing 150 grams, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, a unit package may be furnished. If a unit package is furnished, each can shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve units of issue shall be packaged as specified in 5.2.14.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (See 6.1).

5.3.1 Level A. Eight intermediate packages (96 units) shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Eight intermediate packages (96 units) shall be packed in an exterior container as specified in 5.4.3.6.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

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5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Each intermediate package shall be marking in accordance with paragraph 5.5.3 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legends: "AVOID EXCESSIVE HEAT" and "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F.)"

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legends: "AVOID EXCESSIVE HEAT" and "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-782-6483	TRIAMCINOLONE ACETONIDE AEROSOL, NF, 0.0066%, 150 Grams

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1662

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37497 TRIAMCINOLONE ACETONIDE AEROSOL, NF

...GANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

...PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT

...PARAGRAPH NUMBER AND WORDING.

...MMENDATIONS FOR CORRECTING THE DEFICIENCIES

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